

A Regular Asymptotic Roadmap for SEM Outputs

June 21, 2026

Purpose

This note sketches a paper and implementation roadmap for regular misspecification asymptotics in structural equation modeling. The aim is a math-forward statistical-methods paper, paired with implementation in `magmaan`, that treats many familiar SEM quantities as consequences of a small number of profile-map theorems.

The scope is deliberately regular:

interior pseudo-true points, fixed rank, smooth maps, nonsingular profile Hessians.

Boundary parameters, rank loss, unidentified pseudo-true points, chi-bar-square limits, clipping, and other nonregular phenomena should be acknowledged, but not solved in the first paper. They are too large to include without blurring the central message.

The main claim is not that every mathematical ingredient is new. The claim is that the ingredients are scattered, repeatedly rederived, and often implemented through exact-fit shortcuts. A unified regular theory can make clear which object is being estimated, which Hessian is required under misspecification, and how standard errors, fit measures, nested tests, relative-fit tests, and future indices such as CFI and TLI share the same calculus.

1 Paper Positioning

The paper should be pitched as a statistical-methods contribution with broad SEM impact. A JASA-style target is plausible because the procedures are applied at large scale, the asymptotic objects are nontrivial, and a software-backed unification would be useful beyond one psychometric subcommunity.

The paper should not argue that the work is “too mathematical” for psychometrics journals. That is unnecessary and would read badly. A better framing is:

Routine SEM output is built from profile estimators and profiled discrepancy functions. Under misspecification, the relevant derivatives are observed profile derivatives at pseudo-true points, not the exact-fit/Fisher simplifications often used in derivations and software. We provide a unified regular asymptotic theory and an implementation that applies the same objects to standard errors, model-comparison tests, robust RMSEA, and related fit indices.

This is candid about the contribution: not a new probability limit theorem for its own sake, but a unifying theorem, a correction of common shortcuts, and an implementation-ready framework for procedures that are used constantly.

2 Master Setup

Let $\hat{x}$ be the first-stage object used by the SEM estimator: sample means/covariances, FIML score aggregates, thresholds and polychorics, group proportions, NA-COV components, estimated weights, or any stacked combination needed to make later maps smooth. Assume

$$\hat{x} = x_0 + n^{-1/2}Z + n^{-1}b_n + o_p(n^{-1}), \quad Z \xrightarrow{d} N(0, \Gamma), \quad \mathbb{E}b_n \rightarrow b. \quad (1)$$

The $n^{-1}b$ term is optional for first-order inference but needed for second-order bias corrections such as fixed-misspecification RMSEA.

For each model M , define a smooth criterion

$$f_M(\theta, x), \quad \theta_M(x) = \arg \min_{\theta} f_M(\theta, x), \quad v_M(x) = f_M\{\theta_M(x), x\}.$$

Here v_M is the profiled discrepancy or value function. In a quadratic moment-discrepancy setting,

$$f_M(\theta, x) = \frac{1}{2}r_M(\theta, x)'W(x)r_M(\theta, x),$$

but the theory should be written for a general smooth criterion so that ML, FIML, GLS, ULS, DWLS, ordinal limited-information estimators, and multigroup extensions fit the same template.

All derivatives below are evaluated at (θ_M, x_0) , where $\theta_M = \theta_M(x_0)$ is the pseudo-true parameter. Write

$$H_M = f_{\theta\theta}, \quad G_M = f_{\theta x}.$$

The regularity assumption is that H_M is nonsingular after applying any fixed linear equality constraints or reduced parameterization.

3 Main Theorem 1: Coefficient Maps

The first master map is the pseudo-true parameter map $x \mapsto \theta_M(x)$. The population first-order condition is

$$f_{\theta}\{\theta_M(x), x\} = 0.$$

Differentiating gives

$$D\theta_M[x_0] = -H_M^{-1}G_M.$$

Therefore

$$\sqrt{n}(\hat{\theta}_M - \theta_M) = -H_M^{-1}G_M Z + o_p(1), \quad (2)$$

and

$$\text{Var}\{\sqrt{n}(\hat{\theta}_M - \theta_M)\} = H_M^{-1}G_M \Gamma G_M' H_M^{-1}. \quad (3)$$

This is the master theorem for misspecification-robust standard errors. In SEM notation, H_M is the observed/profile Hessian at the pseudo-true point. Under correct specification it may reduce to expected information or a Gauss–Newton/Fisher expression, but that reduction is a special case, not the general theorem.

4 Main Theorem 2: Residual and Fitted-Value Maps

Many SEM outputs are not just parameters or scalar discrepancies. They use the fitted moments, residuals, standardized residuals, modification indices, EPCs, or score-like projections. These should be handled as smooth maps derived from $\theta_M(x)$.

For a residual map

$$r_M(x) = r_M\{\theta_M(x), x\},$$

the derivative is

$$Dr_M = r_x - r_{\theta} H_M^{-1} G_M, \quad (4)$$

with signs determined by the residual convention. Thus

$$\sqrt{n}\{\hat{r}_M - r_M\} = Dr_M Z + o_p(1).$$

This is the natural home for residual diagnostics, SRMR ingredients, standardized residuals, score-test ingredients, and modification-index building blocks. It is not a new asymptotic primitive, but it deserves a named corollary because many SEM procedures are projection procedures rather than coefficient procedures.

5 Main Theorem 3: Profiled Discrepancy Maps

The second master primitive is the profiled value map

$$v_M(x) = \min_{\theta} f_M(\theta, x).$$

The envelope theorem gives

$$s_M = v_{M,x} = f_x,$$

and the profile Hessian is

$$Q_M = v_{M,xx} = f_{xx} - f_{x\theta} H_M^{-1} f_{\theta x}. \quad (5)$$

Then

$$v_M(\hat{x}) = v_M(x_0) + n^{-1/2} s'_M Z + n^{-1} \left\{ s'_M b_n + \frac{1}{2} Z' Q_M Z \right\} + o_p(n^{-1}). \quad (6)$$

Consequently

$$\mathbb{E}\{v_M(\hat{x})\} = v_M(x_0) + n^{-1} \left\{ s'_M b + \frac{1}{2} \text{tr}(Q_M \Gamma) \right\} + o(n^{-1}), \quad (7)$$

again up to a factor of two depending on whether f_M is defined as F_M or $F_M/2$.

The notation Q_M is important. In the exact-fit, fixed-weight quadratic case, Q_M reduces to the familiar residual projector U_M (up to the same factor convention). Under fixed misspecification, estimated weights, or curved models, Q_M is the correct profile Hessian and U_M is only a shortcut.

6 Contrasts: Nested Tests, Vuong, and Absolute Fit

Most SEM tests compare profiled values. Let

$$c(x) = \sum_{j=1}^J a_j v_{M_j}(x), \quad s_c = \sum_j a_j s_{M_j}, \quad Q_c = \sum_j a_j Q_{M_j}.$$

Then

$$c(\hat{x}) - c(x_0) = n^{-1/2} s'_c Z + \frac{1}{2n} Z' Q_c Z + n^{-1} s'_c b_n + o_p(n^{-1}). \quad (8)$$

Distinguishable regime. If $s_c \neq 0$, the contrast is root- n normal:

$$\sqrt{n}\{c(\hat{x}) - c(x_0)\} \xrightarrow{d} N(0, s'_c \Gamma s_c).$$

This is the regular Vuong-style regime and also the fixed-misspecification regime for many fit-index differences.

Degenerate regime. If the null or comparison makes the values and first derivatives cancel, then the quadratic term is leading:

$$n c(\hat{x}) \xrightarrow{d} \frac{1}{2} Z' Q_c Z,$$

with a weighted chi-square or indefinite quadratic-form reference distribution. Nested difference tests are the main SEM example:

$$c(x) = v_B(x) - v_A(x), \quad Q_c = Q_B - Q_A.$$

Absolute goodness of fit is the special comparison with the saturated model:

$$v_{\text{sat}}(x) \equiv 0, \quad Q_{\text{sat}} = 0, \quad Q_c = Q_M.$$

This explains why a single U_M or Q_M appears in RMSEA and exact-fit tests, while nested tests naturally involve a difference $Q_B - Q_A$.

Local alternatives. Local alternatives belong here too. They perturb $c(x_0)$ or s_c at the n^{-1} or $n^{-1/2}$ scale, turning central quadratic forms into noncentral ones. This should be treated as a regular local regime, distinct from nonregular boundary asymptotics.

7 Fit Indices as Functionals

Fit indices should be treated as smooth or nonsmooth functionals of the master maps. For the first paper, the clean theorem is the smooth one. Let

$$y(x) = \{\theta_{M_1}(x), \dots, \theta_{M_J}(x), r_{M_1}(x), \dots, r_{M_J}(x), v_{M_1}(x), \dots, v_{M_J}(x)\}$$

and let h be differentiable at $y(x_0)$. Then

$$\sqrt{n}\{h(\hat{y}) - h(y_0)\} \xrightarrow{d} N(0, \nabla h' \Sigma_y \nabla h),$$

where Σ_y is assembled from the joint derivatives of the component maps against the common Z .

This covers fixed-positive-misfit versions of:

RMSEA, CFI, TLI, SRMR, information criteria, residual summaries,

provided denominators are nonzero and the functional is evaluated away from kinks. Bias terms can be added by second-order delta method when useful.

The nonsmooth cases should be acknowledged rather than solved in this paper:

$\max(\cdot, 0)$, $\sqrt{\cdot}$ at zero, *clippedCFI/TLI*, *near-zero baselinediscrepancy*, *boundaryparameters*

These are important, but they are not needed for a strong regular-asymptotics paper.

8 Implementation Roadmap in magmaan

The implementation should mirror the theorem structure.

1. Define a first-stage object: $\hat{x}$, x_0 or fitted plug-in, Γ , optional b , and block metadata for moments, weights, groups, and ordinal pieces.
2. For each estimator/model, expose a profile object that can compute f_M , $\hat{\theta}_M$, H_M , G_M , residual derivatives, s_M , and Q_M .
3. Route standard errors through the coefficient theorem (3).
4. Route residual diagnostics, score tests, modification indices, and SRMR ingredients through the residual-map theorem (4).

5. Route exact-fit tests, nested tests, Vuong tests, RMSEA corrections, and future CFI/TLI corrections through the value-map theorem (6).
6. Keep U_M as an optimized special case for exact-fit/fixed-weight quadratic models, but store and test the general object as Q_M .

Validation should proceed from the simplest profile maps to the most coupled:

1. fixed-weight ULS/GLS, complete data;
2. estimated-weight GLS/DWLS and ordinal limited information;
3. ML/FIML as criterion-specific profile maps;
4. multigroup and equality-constrained comparisons;
5. fit-index functionals such as RMSEA, CFI, TLI, and SRMR.

9 Proposed Paper Structure

1. Introduction: widely used SEM outputs rely on exact-fit shortcuts; under misspecification, one needs profile derivatives at pseudo-true points.
2. Setup: first-stage CLT, smooth criteria, regular pseudo-true points.
3. Main theory: coefficient maps, residual maps, value maps.
4. Model comparisons: absolute fit, nested tests, Vuong and overlapping comparisons, local alternatives.
5. Fit indices: RMSEA now; CFI/TLI/SRMR as smooth functionals, with nonsmooth cases flagged.
6. SEM specializations: ML/FIML, GLS/ULS/WLS/DWLS, ordinal/multigroup.
7. Implementation: `magmaan` architecture, analytic derivatives, finite-difference validation, lavaan parity where relevant.
8. Simulations and examples: cases where exact-fit shortcuts agree, cases where Q_M and observed-profile Hessians matter, and empirical size/coverage for robust SEs and tests.
9. Discussion: regular theory first; boundaries, rank loss, and clipped fit indices are future work.

10 Immediate Next Steps

The next concrete work should be:

1. Write a short theorem note with consistent notation $(x, \Gamma, b, \theta_M, v_M, s_M, Q_M, H_M, G_M)$.
2. Refactor existing RMSEA and nested-test notes to cite that notation rather than each defining a parallel vocabulary.
3. Add a small fixed-weight validation script showing $n\{\mathbb{E}v_M(\hat{x}) - v_M(x_0)\}$ matches $\frac{1}{2} \text{tr}(Q_M \Gamma)$ under fixed misspecification.
4. Add an estimated-weight validation that isolates the weight channel.
5. Decide the first implemented public surface in **magmaan**: likely profile Hessians Q_M , robust nested tests, and fixed-misspecification RMSEA diagnostics before moving to CFI/TLI.